1.07 Algebra: Simultaneous Equations

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Solutions sets

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The solution(s) to an equation may be:

A single value:	$2x + 1 = 5$
Multiple values:	$x^2 + 3x + 2 = 0$
An infinitely large set of values:	$x > 3$
No (real) values!	$x^2 = -1$
Every value!	$x^2 + x = x(x + 1)$

The point is that you shouldn't think of the solution to an equation/inequality as an 'answer', but a **set** of values, which might just be a set of 1 value (known as a singleton set), a set of no values (i.e. the empty set $\varnothing$), or an infinite set (in the last example above, this was $\mathbb{R}$)

✎ The solutions to an equation are known as the **solution set**.

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Solutions sets

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For simultaneous equations, the same is true, except each 'solution' in the solution set is an assignment to **multiple** variables.
All equations have to be satisfied **at the same time, i.e. 'simultaneously'**.

Scenario	Example	Solution Set
A single solution:		
Two solutions:		
No solutions:		
Infinitely large set of solutions:		

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Recap!

Solve the simultaneous equations:

$3x + y = 8$
$2x - 3y = 9$

We can either use substitution (i.e. making x or y the subject of one equation, and substituting it into the other) or elimination, but the latter is easier for linear equations.

$\mathbf{9x+3y=24}$

Solve the simultaneous equations:

$x + 2y = 3$
$x^2 + 3xy = 10$

We can't use elimination this time as nothing would cancel.
We instead:
(1) Rearrange linear equation to make x or y the subject.
(2) Substitute into quadratic equation and solve.

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Test Your Understanding

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Solve the simultaneous equations:

$3x^2 + y^2 = 21$
$y = x + 1$

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Exercise 3A/B – Y1 Pages 40, 41

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4 Solve the simultaneous equations:
$x + y = 3, \quad x^2 - 3y = 1$

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5 a By eliminating y from the equations
$y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, show that $x^2 - 2x - 11 = 0$. **b** Hence, or otherwise, solve the simultaneous equations
$y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, giving your answers in the form
$a \pm b\sqrt{3}$, where a, b are integers.

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6 One pair of solutions for the simultaneous equations $y = kx - 5, \quad 4x^2 - xy = 6$ is $(1, p)$ where k, p ar

a Find the values of k, p.

b Find the second pair of solutions for the simultaneous equations.

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Extension

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1 *[MAT 2012 1G]* There are *positive* real numbers x and y which solve the equations
$2x + ky = 4,$
$x + y = k$
for:
A) All values of k;
B) No values of k;
C) $k = 2$ only;
D) Only $k > -2$

If $\mathbf{k=2}$ then $\mathbf{2x+2y=4}$ and $\mathbf{x+y=2}$ which are equivalent.

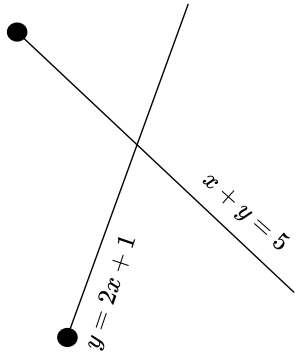
2 *[STEP 2010 Q1]* Given that
$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$
a) Find the values of a, b, c, d.
b) Solve the simultaneous equations:
$5x^2 + 2y^2 - 6xy + 4x - 4y = 9,$
$6x^2 + 3y^2 - 8xy + 8x - 8y = 14$
(Hint: Can we use the same method in (a) to rewrite the second equation?)

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Simultaneous Equations and Graphs

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Recall that a line with a given equation is **the set of all points that satisfy the equation**.

i.e. It is a graphical representation of the solution set where each (x, y) point represents each of the solutions x and y to the equation.
e.g. $x = 3, y = 7$

Now suppose we introduced a second simultaneous equation:
$y = 2x + 1$
$x + y = 5$

The second line again consists of all points (x, y) which satisfy the equation. So what point must satisfy both equations simultaneously?
The point of intersection!

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Example

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a) On the same axes, draw the graphs of:
$2x + y = 3$
$y = x^2 - 3x + 1$

b) Use your graph to write down the solutions to the simultaneous equations.

c) What algebraic method (perhaps thinking about the previous chapter), could we have used to show the graphs would have intersected twice?

Substituting linear equation into quadratic:
$\mathbf{y = 3 - 2x}$

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$a = 1, b = -1, c = -2$

$1 + 8 = 9 > 0$

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Thus the quadratic has two distinct solutions,

Another Example

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a) On the same axes, draw the graphs of:
$y = 2x - 2$
$y = x^2 + 4x + 1$

b) Prove algebraically that the lines never meet.

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Final Example

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The line with equation $y = 2x + 1$ meets the curve with equation $kx^2 + 2y + (k - 2) = 0$ at exactly one point. Given that k is a positive constant:
a) Find the value of k.
b) For this value of k, find the coordinates of this point of intersection.

a

$k = 2 \quad 2x \quad 4x \quad 2 = 0$

b

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Exercise 3C – Y1 Page 45

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8 Given the simultaneous equations $2x - y = 1$, $x^2 + 4ky + 5k = 0$, where k is a non-zero constant,

a show that $x^2 + 8kx + k = 0$.

b Given that $x^2 + 8kx + k = 0$ has equal roots,
find the value of k.

c for this value of k, find the solution of the simultaneous equations.

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9 A swimmer dives into a pool. Her position, p m, underwater can be modelled in relation to her horizontal distance, x m, from the point she entered the water as a quadratic equation $p = \frac{1}{2}x^2 - 3x$. The position of the bottom of the pool can be modelled by the linear equation $p = 0.3x - 6$. Determine whether this model predicts that the swimmer will touch the bottom of the pool.

